TRESATA WRITTEN BRIEF: AI-ASSISTED COVENANT COMPLIANCE ENGINE

System Architecture, Mathematical Formalism, Empirical SEC Verification & Engineering Report

Focus: Quarterly Maintenance Total Net Leverage Ratio Compliance | **Author:** AI Agent & Lead Architect | **Date:** November 2026

1.1 Executive Summary

In corporate leveraged finance and syndicated lending, credit agreements govern billions of dollars in enterprise debt through restrictive financial covenants. Among these covenants, the **quarterly maintenance Total Net Leverage Ratio** (Consolidated Net Debt / LTM Consolidated EBITDA) is the single most critical solvency test. A covenant breach triggers immediate technical default, accelerating debt maturity and shifting corporate control to lenders. Historically, compliance certification has been a high-friction, error-prone manual process where credit analysts spend hundreds of hours reconciling 200+ page contracts, tracking multi-year amendments, mapping general ledger accounts, and arbitrating borrower-claimed EBITDA add-backs against bespoke contractual caps.

The Fatal Flaws of Pure LLMs in Credit Operations: Deploying generic LLM agents directly on covenant math leads to catastrophic operational failures: (1) *Arithmetic Hallucination:* LLMs cannot reliably perform nested numerical cap mathematics, particularly circular post-addback caps where an add-back is limited to a percentage of EBITDA calculated *after* giving effect to the add-back itself; (2) *Citation Drift & Temporal Invention:* LLMs routinely invent clause references, assume unstated effective dates, or hallucinate page numbers in SEC HTML filings; (3) *Arbitrary Arbitration:* When multiple expenses exceed an aggregate cap without an explicit contractual waterfall, LLMs arbitrarily pick an expense to disallow rather than halting for human review.

1.2 Problem Formulation & Architectural Philosophy

To eliminate these failure modes while maximizing automation efficiency, we architected an enterprise compliance system based on three non-negotiable principles:

• **Division of Cognitive Labor:** The LLM functions strictly as a semantic clause parser and qualitative evidence auditor. Python is the 100% deterministic mathematical authority—zero arithmetic is ever delegated to an LLM.
• **Contractual Provenance & Cryptographic Integrity:** Every document is hashed (SHA-256). All internal sections and HTML paragraph blocks are assigned stable deterministic identifiers (InternalRef: internal_block_X).
• **Zero Guessing & Mandatory Human-in-the-Loop (HITL) Gates:** Ambiguous contractual language, non-positive EBITDA, or unallocated aggregate cap overruns strictly halt calculation with **Cannot compute**, requiring credit officer resolution.

PHASE 1: CREDIT AGREEMENT & AMENDMENT DIGITIZATION	PHASE 2: QUARTERLY COMPLIANCE CHECKING ENGINE
• **Document Ingestion:** SEC EDGAR HTML/PDF with SHA-256 checksums. • **Amendment Tracker:** Extracts deltas (REPLACE, INSERT, DELETE). • **Agreement Timeline:** Chronological, immutable contract versioning. • **Dependency Graph:** Recursive BFS DAG traversal from covenant roots. • **Rule Set Synthesis:** Verbatim quotation and mathematical boundary checks. • **[GATE 1: Rule Set Sign-off]:** Mandatory credit officer approval.	• **Financial Data Ingestion:** Multi-period IS, BS, Ledger, Claimed Add-backs. • **Account Mapper:** Maps general ledger to covenant categories. • **[GATE 2: Account Mapping Sign-off]:** Reviewer validates ledger buckets. • **Add-back Qualitative Auditor:** LLM classifies (ELIGIBLE / INELIGIBLE). • **[GATE 3: Audit Sign-off]:** Credit officer reviews and overrides adjustments. • **100% Deterministic Python Math:** Caps, circular algebra, netting, ratios.

1.3 Core Covenant Metrics & Formal Definitions

The system evaluates the Total Net Leverage Ratio against the contractual ceiling: **Total Net Leverage Ratio = Consolidated Net Debt / LTM Consolidated EBITDA**. Consolidated Net Debt is defined as Funded Debt minus unrestricted cash, subject to contractual cash netting caps (e.g., $37.5M in Whole Earth Brands). LTM Consolidated EBITDA is computed over a rolling 4-quarter period, incorporating eligible borrower add-backs subject to individual category caps, circular post-EBITDA bounds, and aggregate basket ceilings.

2. SYSTEM ARCHITECTURE & TECHNICAL SPECIFICATIONS

2.1 End-to-End Architectural Workflow

The diagram below details the dual-phase pipeline from raw SEC document ingestion to auditable compliance certification:

PIPELINE STAGE	TECHNICAL MECHANISM & ALGORITHMIC SAFEGUARDS	GOVERNANCE GATE
1. Ingestion	Parses SEC EDGAR HTML/PDF into atomic content blocks with SHA-256 hashes. Assigns immutable InternalRef: internal_block_X identifiers. Prevents hallucinated legal paragraph numbers.	Cryptographic Checksum Verification
2. Amendment Tracking	Identifies deltas (REPLACE, INSERT, DELETE). Python verifies supporting verbatim quotes against base document. Maintains chronological timeline of immutable AgreementVersion snapshots.	Chronological Date Check & Verbatim Quote Match
3. Dependency DAG	Builds Directed Acyclic Graph via recursive BFS traversal starting strictly from covenant root terms. Prunes non-financial boilerplate and flags circular post-EBITDA cap references for algebraic resolution.	Circularity & Missing Term Detection
4. Rule Set Synthesis	Extracts EBITDA additions, deductions, caps, and covenant ratio stepping schedules. Python engine verifies that all extracted clauses match contract text verbatim.	**[GATE 1] Mandatory Rule Set Sign-off by Credit Officer**
5. Account Mapping	Ingests multi-period Trial Balance / General Ledger. Maps account codes to covenant categories with confidence scoring. Flags ambiguous ledgers for human approval.	**[GATE 2] Reviewer Approves Ledger Mappings**
6. Add-Back Audit	LLM audits claimed expenses against contractual criteria, classifying items as ELIGIBLE, INELIGIBLE, or NEEDS_HUMAN_REVIEW. Reviewer can override classifications.	**[GATE 3] Reviewer Approves / Overrides Add-backs**
7. Deterministic Math	100% Python arithmetic. Solves circular post-EBITDA caps algebraically. Applies cash netting ceilings. Computes ratio, headroom, and cushion percentage across multiple readings.	**Deterministic Output: 4-Case Status & Audit Workpaper**

2.2 Core Technical Innovations

1. Closed-Form Algebraic Resolution of Circular Add-back Caps:
Credit agreements routinely cap restructuring or synergy add-backs to a percentage of Consolidated EBITDA calculated *after* giving effect to the add-back (e.g. Redwire Corporation Section 1.01 limits synergy add-backs to 25% of post-addback EBITDA). Formally: $\mathbf{A \le x \cdot (B + A)}$, where *A* is the capped add-back, *B* is baseline uncapped EBITDA, and *x* is the cap percentage. Rearranging: $\mathbf{A \cdot (1 - x) \le x \cdot B \Rightarrow A \le [x / (1 - x)] \cdot B}$. For a 25% cap (x = 0.25): **Max Add-back = (0.25 / 0.75) · B = (1/3) · B ≈ 33.333% of Baseline EBITDA**. Rather than iterative numerical approximation, Python solves this in exact closed form.

2. Contractual Cash Netting Caps:
While standard models subtract all unrestricted cash, leveraged agreements frequently cap cash netting (e.g., Whole Earth Brands Section 1.01 limits netted cash to exactly **$37,500,000**). Our calculation engine computes: **Consolidated Net Debt = Funded Debt - min(Unrestricted Cash, Cash Netting Cap)**.

3. The 4-Case Compliance Status Taxonomy:
To prevent misleading binary pass/fail claims under contractual ambiguity, the engine outputs four distinct statuses: (1) **Pass under all readings**: Ratio ≤ covenant ceiling across primary and alternative readings; (2) **Fail under all readings**: Ratio > covenant ceiling across all readings; (3) **Outcome depends on a reading**: Compliant under one interpretation but failing under another (flags exact variance for credit committee); (4) **Cannot compute**: Strictly halted due to missing data, non-positive EBITDA, unapproved gates, or unallocated aggregate cap overruns.

4. Mathematical Headroom & Cushion Formulas:
- **Ratio Headroom:** Covenant Ceiling - Actual Total Net Leverage Ratio
- **Absolute EBITDA Buffer ($):** Actual LTM EBITDA - (Consolidated Net Debt / Covenant Ceiling)
- **EBITDA Cushion (%):** (Absolute EBITDA Buffer / Actual LTM EBITDA) × 100% *(represents the percentage drop in EBITDA triggering technical breach).*

3. EMPIRICAL VALIDATION & TEST RESULTS

3.1 Empirical Verification Against Real SEC EDGAR Filings

The system was verified against 8 official SEC EDGAR filings across three public companies maintaining quarterly Total Net Leverage Ratio covenants:

COMPANY	DOCUMENT & FORM	EFFECTIVE	SHA-256 CHECK (12 CHARS)	VERIFIED CONTRACTUAL MECHANICS
Redwire Corp	Credit Agreement (10-K Ex-10.13)	2020-10-28	3a5a70bc5020...	Section 7.11 ratio schedule (5.00x → 4.50x); 25% circular post-EBITDA cap.
Redwire Corp	Fifth Amendment (8-K Ex-10.1)	2022-10-28	f5a48c8303a7...	Replaced Section 7.11(a) schedule; stepped ratio ceiling up to 6.50:1.00.
Redwire Corp	Sixth Amendment (8-K Ex-10.1)	2023-06-23	44cd8c96d528...	Replaced Section 7.11(a) schedule; stepped ratio ceiling up to 7.50:1.00.
Whole Earth	A&R; Agreement (8-K Ex-10.1)	2021-02-05	68df9acc9b5b...	Section 7.8 ratio (4.00x → 5.50x); **$37.5M cash netting cap**; $10M / 12.5% fee cap.
Whole Earth	First Amendment (8-K Ex-10.1)	2022-06-15	9914760a8c0a...	Inserted fixed deemed quarterly EBITDA run-rates; revolving facility resize.
Mercury Systems	Credit Agreement (8-K Ex-10.1)	2016-05-02	73e51554c19d...	Section 8.11(b) Total Net Leverage Ratio; uncapped cash netting.
Mercury Systems	Amendment No. 5 (8-K Ex-10.1)	2023-11-07	d07b1bf5562f...	Modified covenant relief period leverage thresholds.
Mercury Systems	Amendment No. 6 (8-K Ex-10.1)	2024-08-12	eb7bcfa77dce...	Ratio 5.25x → 4.50x; +0.50x acquisition holiday; EAC Adjustments in clause (x)(II).

3.2 Empirical Results Across 12 Synthetic Stress Scenarios (54 Passed, 1 Skipped)

All 12 synthetic test cases passed exact numerical thresholds and status requirements in automated testing (54 passed, 1 skipped in pytest):

#	SCENARIO DESCRIPTION	KEY MECHANISM & CAPS	EXPECTED STATUS	ACTUAL STATUS	LEVERAGE	CUSHION %
1	Comfortably Compliant	Low debt ($130M Net Debt / $125M EBITDA)	Pass under all readings	**Pass under all readings** ■	1.0400x	76.89%
2	Direct Covenant Breach	High debt ($445M Net Debt / $60M EBITDA vs 4.50x)	Fail under all readings	**Fail under all readings** ■	7.4167x	-64.81%
3	Capped Add-back Haircut	$25M claimed restructuring restricted to $6.67M cap	Pass under all readings	**Pass under all readings** ■	1.6721x	62.84%
4	Circular Post-EBITDA Cap	Exact algebra: (0.25/0.75)*$30M = $10.0M limit	Pass under all readings	**Pass under all readings** ■	1.4138x	68.58%
5	Contractual Cash Netting Cap	$60M cash capped at $37.5M ceiling ($22.5M unnetted)	Pass under all readings	**Pass under all readings** ■	2.0312x	54.86%
6	Unallocated Cap Overrun	Two categories exceed $6.67M shared basket without waterfall	Cannot compute	**Cannot compute** ■	N/A	N/A
7	Outcome Depends on Reading	Reading A: 4.500x (Pass); Reading B: 4.615x (Fail)	Outcome depends on reading	**Outcome depends...** ■	4.5000x	1.10%
8	Negative EBITDA Scenario	LTM EBITDA -$55M; prevents ratio calculation; review flag	Cannot compute	**Cannot compute** ■	N/A	N/A
9	Zero EBITDA Scenario	LTM EBITDA exactly $0.00; ZeroDivisionError prevented	Cannot compute	**Cannot compute** ■	N/A	N/A
10	Missing Approvals Gate	Unapproved RuleSet strictly halts computation	Cannot compute	**Cannot compute** ■	N/A	N/A
11	Ineligible Add-Back	Routine capex audited as INELIGIBLE receiving $0 credit	Pass under all readings	**Pass under all readings** ■	1.6279x	63.82%
12	Unit Multiplier Scaling	Financial statements reported in $000s scaled accurately	Pass under all readings	**Pass under all readings** ■	1.6279x	63.82%

3.3 Operational Limitations & Production Boundaries

1. **Shared Basket Overruns Require Human Allocation:** If an agreement fails to specify an allocation waterfall for an aggregate cap overrun, the engine halts with Cannot compute until a credit analyst records a haircut allocation.
2. **Live OpenRouter Disclosure & Offline Fallback:** In live testing, the OpenRouter connection test verified successfully (HTTP 200 'PONG'). However, actual live Rule Set synthesis with free models encountered upstream model token/content limits on large legal texts (returning null/empty content). As designed by the dual-engine architecture, the application safely reverted to the verified deterministic contract template (offline fallback). The system does not claim live AI generation succeeded in production.
3. **Unstructured ASCII SEC Tables:** SEC HTML filings with non-standard ASCII text tables may require analyst review during ingestion.

4. APPENDIX — USER CONTRIBUTION & ENGINEERING LEADERSHIP

4.1 Collaborative Pair-Programming Overview

This project was designed, implemented, and verified through an intensive collaborative pair-programming engagement between the **USER (acting as Lead Financial Architect & Systems Director)** and the **AI Assistant (Antigravity, acting as Senior Implementation Engineer)**. The User provided authoritative credit domain expertise and architectural governance, steering the system away from generic LLM pitfalls toward an institutionally rigorous, auditable compliance platform.

4.2 Key Architectural Decisions Directed by the User

1. **Rejection of Non-Matching Covenant Ratios:** Discarded Hertz, AMC, and Six Flags when SEC filings showed First Lien / Secured covenants rather than Total Net Leverage Ratio. Selected Redwire, Whole Earth Brands, and Mercury Systems.
2. **The Zero-Guessing Mandate:** Strictly prohibited LLM guessing or automatic arbitration. Any unverified quote or unallocated basket overrun must pause for human review and return Cannot compute.
3. **Algebraic Resolution of Circular Caps:** Formulated circular post-addback caps as closed-form algebraic constraints ($A \le [x/(1-x)] \cdot B$) rather than lossy numerical iterations.
4. **Contractual Cash Netting Ceiling:** Mandated support for capped cash netting (e.g. Whole Earth Brands' $37.5M ceiling) distinct from uncapped agreements.
5. **No Automatic Default on Negative EBITDA:** Required that negative or zero EBITDA halt computation with a review flag rather than declaring an automatic covenant breach.
6. **EBITDA Cushion Definition:** Formulated EBITDA Cushion (%) strictly as the percentage drop in EBITDA that would trigger a breach: Cushion % = (Absolute Buffer / Actual EBITDA) × 100%.
7. **Elimination of Invented Effective Dates:** Eliminated fallback dates, requiring contract preamble extraction or explicit reviewer validation.
8. **Internal Reference Citation Standard:** Mandated that parsed HTML blocks be cited as internal references (InternalRef: internal_block_X) rather than inventing fake legal paragraph numbers.

4.3 Engineering Governance Matrix

PHASE & DOMAIN	USER STRATEGIC GOVERNANCE & ARCHITECTURAL DIRECTIVES	AI TECHNICAL EXECUTION & CODEBASE IMPLEMENTATION
Phase 1: Ingestion & Amendments	Defined provenance fidelity rules; prohibited fake HTML page numbers; mandated SHA-256 checks and chronologically ordered timelines.	Built ingestion.py and amendment_tracker.py; verified 8 SEC filings; implemented AgreementTimeline snapshot engine.
Phase 1: Dependency Graph	Directed recursive BFS traversal starting strictly from covenant roots; required cycle and missing term detection.	Implemented dependency_graph.py; pruned non-financial boilerplate; classified circular EBITDA cap references.
Phase 1: Rule Set Synthesis	Mandated verbatim quotation checks, ambiguity preservation, and Gate 1 human approval before financial processing.	Built ruleset_generator.py; integrated OpenRouter with deterministic fallback; verified math bounds and quote accuracy.
Phase 2: Financial Ingestion	Defined quarterly deliverable schema, ledger account mapping, and Gate 2 reviewer sign-off.	Built financial_data.py and account_mapper.py with confidence scoring and manual override flow.
Phase 2: Add-Back Auditing	Required qualitative classification (ELIGIBLE, INELIGIBLE, NEEDS_REVIEW) and Gate 3 reviewer overrides.	Built addback_auditor.py with OpenRouter prompts and deterministic keyword auditor.
Phase 2: Math Engine & Status	Formulated 4-case compliance taxonomy; prohibited automatic aggregate overrun arbitration; specified headroom and cushion formulas.	Built calculation_engine.py; implemented closed-form cap algebra, cash netting limits, and JSON/Markdown workpaper export.
Testing & UI Delivery	Specified at least 10 synthetic test cases; mandated live/offline API status transparency and standalone Streamlit interface.	Built 12 synthetic cases; built Streamlit web interface (app.py); verified all 54 pytest test cases pass cleanly (1 skipped for live key).

SYSTEM STATUS & SUBMISSION READINESS: The complete codebase, documentation, verified SEC EDGAR credit agreements, and interactive Streamlit web application have been fully tested (54 passed, 1 skipped) and are ready for institutional deployment.